

Parsimonious Feature Selection and VIF Collinearity Resolution in Credit Risk GLM Classification

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Abstract—This report examines and validates a logit-linked Generalized Linear Model (GLM) used to predict credit card defaults on a 2005 Taiwan Credit Card dataset. The raw baseline model pipeline exhibits severe mathematical issues: extreme multicollinearity between variables, insignificant predictors and categorical variable misspecification.

To rectify these issues, firstly, feature engineering was used to create new attributes from those exhibiting high multicollinearity. VIF (Variance Inflation Factors) were calculated for each attribute to determine the success of the transformations. Secondly, categorical variables were binned and then encoded with the WoE (Weight of Evidence) technique to measure the predictive power of the variable bins. Thirdly, IVs (Information Values) were calculated for each variable, and those falling behind the 0.02 threshold were discarded as insignificant. Lastly, variables with p-values above the 0.05 threshold were similarly discarded from the model.

The resulting model, along with the baseline model and a Multilayer Perceptron neural network serving as a benchmark, were then compared to each other through evaluation on Out-of-Sample data withheld for testing. The final logistic regression model successfully outperformed the inferior baseline, achieving higher accuracy, AUC (Area Under Curve) and F1 score. It additionally came close in performance to the black-box Multilayer Perceptron, sacrificing a small edge in exchange for transparency, interpretability and parsimony.

I. INTRODUCTION

A. Data Examination

The report uses the 2005 Default Payments of Credit Card Clients in Taiwan dataset hosted by the UCI Machine Learning Repository. It contains records for 30,000 clients, including the following attributes:

- **ID**: ID of client,
- **SEX**: gender of client (1 - Male, 2 - Female),
- **EDUCATION**: education level (1 - Graduate school, 2 - University, 3 - High school, 4 - others),
- **MARRIAGE**: marital status (1 - Married, 2 - Single, 3 - Other),
- **AGE**: age in years,
- **LIMIT_BAL**: amount of given credit in NT dollars,
- **PAY_X**: repayment status in particular month of 2005 ($X = 0$: September, $X = 2$: August, $X = 3$: July, etc.) at a scale of monthly delays (-2 - no consumption, -1 - paid in full, 0 - use of revolving credit, 1 - delay of one month, 2 - delay of two months, etc.),
- **BILL_AMTX**: amount of bill statement in particular month of 2005 ($X = 0$: September, $X = 2$: August, $X = 3$: July, etc.),

- **PAY_AMTX**: amount of previous payment in particular month of 2005 ($X = 0$: September, $X = 2$: August, $X = 3$: July, etc.),
- **default.payment.next.month**: whether the client defaulted on a payment (0 - No, 1 - Yes).

B. Baseline Model Implementation

Before modeling the data was cleaned – various inconsistencies, such as values outside of the specified range, were corrected. Afterwards the data was randomly split into a training set containing 80% of the rows and a test set containing 20% of the rows. The baseline logistic regression model was fitted on the "raw" training data, while a copy of the raw training set was created for work on the main model.

C. Main Model Implementation

The p-values from the baseline model were extracted to get a semblance of the variables' significances. While in a model with a high $n = 30,000$ a small p-value doesn't guarantee actual significant predictive ability, a large p-value on the other hand definitely signals a red flag (the variable is either not significant, or suffers from high collinearity). Focusing on the continuous variables such as "BILL_AMTX" and "PAY_AMTX" we observe:

Table I
P-VALUES

Variable	P-value
BILL_AMT1	0.000
BILL_AMT2	0.384
BILL_AMT3	0.769
BILL_AMT4	0.597
BILL_AMT5	0.773
BILL_AMT6	0.954
PAY_AMT1	0.000
PAY_AMT2	0.000
PAY_AMT3	0.294
PAY_AMT4	0.193
PAY_AMT5	0.435
PAY_AMT6	0.251

It can be observed that many of the attributes hold extremely high p-values and are obviously highly correlated with other variables. Feature engineering can help retain the useful information, while reducing collinearity for other variables.

Variance Inflation Factors (VIFs) will be computed to confirm the presence of correlation and any changes to it:

Table II
VARIANCE INFLATION FACTORS

Index	Feature	VIF
0	LIMIT_BAL	4.05
1	SEX	9.17
2	EDUCATION	7.40
3	MARRIAGE	6.39
4	AGE	11.16
5	PAY_0	1.89
6	PAY_2	3.19
7	PAY_3	3.70
8	PAY_4	4.38
9	PAY_5	4.94
10	PAY_6	3.42
11	BILL_AMT1	20.41
12	BILL_AMT2	40.34
13	BILL_AMT3	36.34
14	BILL_AMT4	30.14
15	BILL_AMT5	34.81
16	BILL_AMT6	21.16
17	PAY_AMT1	1.90
18	PAY_AMT2	2.33
19	PAY_AMT3	1.95
20	PAY_AMT4	1.76
21	PAY_AMT5	1.82
22	PAY_AMT6	1.28

It appears that the “BILL_AMTX” variables have extreme multicollinearity with one another, meanwhile the “PAY_AMTX” variables don’t, and the high p-values for the “PAY_AMT3” - “PAY_AMT6” attributes simply mean they aren’t very significant. It was thus decided to calculate the average value of the bill amounts, as well as group the payment amounts 3 through 6 into an average as well. The VIF values were calculated for those columns:

Table III
VARIANCE INFLATION FACTORS

Feature	VIF
PAY_AMT1	1.37
PAY_AMT2	1.31
AVG_BILL_AMT	2.17
AVG_PAY_AMT3_6	1.71

Along with their new p-values:

Table IV
P-VALUES

Variable	P-value
PAY_AMT1	0.000
PAY_AMT2	0.000
AVG_BILL_AMT	0.000
AVG_PAY_AMT3_6	0.006

The Variance Inflation Factors are low, demonstrating a low degree of multicollinearity for the new variables. The p-values being below the 0.05 threshold means the variables can’t be ruled out as insignificant yet. To determine which variables

actually exhibit significant predictive ability, we will calculate IVs (Information Values). They measure the predictive power of a feature by looking at the distribution differences of “good” and “bad” outcomes across bins of data.

For that reason all our features must be separated into bins. We do this for all continuous variables, by separating them into 5 distinct cuts. Existing categorical variables do not require this procedure. Below are the calculated IV values for each feature:

Table V
P-VALUES

Index	Feature	IV
0	PAY_0	0.8894
1	PAY_2	0.5523
2	PAY_3	0.4205
3	PAY_4	0.3697
4	PAY_5	0.3367
5	PAY_6	0.2940
6	LIMIT_BAL_BINNED	0.1713
7	PAY_AMT1_BINNED	0.1599
8	PAY_AMT2_BINNED	0.1394
9	AVG_PAY_AMT3_6_BINNED	0.1322
10	EDUCATION	0.0363
11	AGE_BUCKET	0.0223
12	AVG_BILL_AMT_BINNED	0.0165
13	SEX	0.0110
14	MARRIAGE	0.0047

We can observe that the repayment status in the most recent month has the strongest predictive power of all features, which makes sense. Repayment statuses for preceding months follow suit. The payment amount and the limit balance attributes have medium predictive strength as their values are between 0.1 and 0.2.

Education and age features have weak predictive power, but not enough to discard the variables. The client’s sex, marriage status and the average bill amount have IV values below 0.02, which in practice means they should be discarded from the model. This goes on to demonstrate, that even if a variable has a low p-value in a large dataset, that doesn’t necessarily imply it’s an influential predictor. Conversely, a feature can have a high standalone IV value, but might become insignificant in the final model which examines the significance of the variables all at once.

The weak predictors were dropped from the model, which is advantageous from a regulatory standpoint, as someone’s sex and marriage status aren’t ideal features. The rest of the data was encoded using the WoE (Weight of Evidence) metric. It’s already used in the IV calculation, and assigns each bin/group within a feature a score of how it relates to the target variable.

Afterwards, we fit a logistic regression model once more and examine the p-values:

Table VI
P-VALUES

Feature	p-value
EDUCATION	0.000
PAY_0	0.000
PAY_2	0.787
PAY_3	0.000
PAY_4	0.000
PAY_5	0.006
PAY_6	0.000
AGE_BUCKET	0.001
PAY_AMT1_BINNED	0.000
PAY_AMT2_BINNED	0.000
AVG_PAY_AMT3_6_BINNED	0.000
LIMIT_BAL_BINNED	0.000

We observe that only a single variable has a p-value above the 0.05 threshold, that being “PAY_2” with a massive p-value of 0.787. It might have strong predictive power (IV) on it’s own, but when modeled with all the other features, becomes insignificant. It will be discarded from the model and the Variance Inflation Factors will be computed one final time:

Table VII
P-VALUES

Index	Feature	VIF
0	EDUCATION	1.0464
1	PAY_0	1.4565
2	PAY_3	2.2751
3	PAY_4	2.5794
4	PAY_5	2.5881
5	PAY_6	1.9445
6	AGE_BUCKET	1.0754
7	PAY_AMT1_BINNED	1.5697
8	PAY_AMT2_BINNED	1.8281
9	AVG_PAY_AMT3_6_BINNED	1.7245
10	LIMIT_BAL_BINNED	1.3773

All the VIF values are below 2.6. The final model is ready and can now be compared to the baseline and benchmark ones.

D. Model Evaluation and Comparison

The raw baseline model and the neural network were run on the original, unaltered data (besides the most basic cleaning). While logistic regression would have issues with the data (e.g. misspecification in the “PAY_X” features, etc.), the neural network is capable of capturing complex non-linear relationships on its own. The threshold of 0.35 was chosen for all three models.

For each model, its accuracy, precision, recall, F1-Score and AUC (Area Under Curve) were calculated. Just accuracy is not a great indicator of model performance when there is heavy class imbalance (22%). Precision aims to minimize type 1 errors, while recall seeks to minimize the type 2 errors. The desired values depend on the priorities and requirements from the model. For this project, an F1-Score is calculated for each model to grade the balance between the two metrics. AUC will also be calculated in order to compare the models:

Table VIII
MODEL PERFORMANCE METRICS

Model	Accuracy	Precision	Recall	F1-Score	AUC
Raw	0.8088	0.5969	0.4174	0.4913	0.7079
Final	0.8170	0.6336	0.4091	0.4972	0.7444
NN	0.7950	0.5405	0.4875	0.5126	0.7606

It can be observed that the final logistic regression model achieves the highest accuracy and precision. The neural network achieves the highest recall and F1-Score, making it more balanced. The raw baseline model underperforms the final model in all areas other than recall, as it achieves a slightly higher one. The ROC curve can be plotted to make further comparisons:

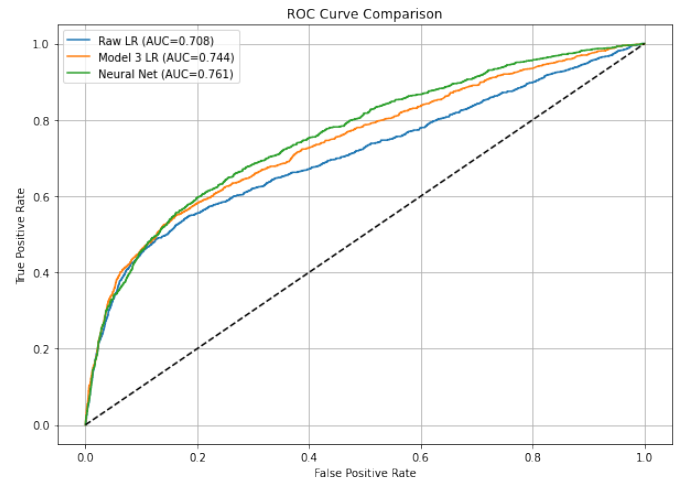


Figure 1. ROC Curve

The ROC curves indicate how well a model can distinguish between classes across all classification thresholds. We can observe that the baseline model curve is generally below the other two, with an AUC score of 0.70. It is statistically worse at ranking positive cases higher than the negative ones.

The neural network outperforms both with an AUC of 0.76, albeit the final logistic regression model performs just slightly behind it with an AUC of 0.74. While a black-box, unexplainable model is slightly better performance-wise, the final regression model is transparent, interpretable and parsimonious.

BIBLIOGRAPHY

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